

MATH Unit 4 — Discrete Mathematics & Logic

Complete Notes

Mathematics (MATH) · Unit 4 · Discrete Mathematics & Logic · 2025

Overview

These notes cover **Discrete Mathematics & Logic**. Engineering and discrete mathematics — linear algebra, calculus, probability, logic and graph theory for computer science. Each section states the core idea, a worked intuition and the points examiners look for.

1. Propositional Logic

Propositions combine with connectives; truth tables test tautologies and contradictions.

Why it matters: Propositional Logic appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of propositional logic in one line.
- A small example showing propositional logic in practice.
- Common mistakes students make and how to avoid them.

2. Predicate Logic

Quantifiers for all and there exists extend propositional logic to statements about objects.

Why it matters: Predicate Logic appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of predicate logic in one line.
- A small example showing predicate logic in practice.
- Common mistakes students make and how to avoid them.

3. Rules of Inference

Modus ponens, modus tollens and resolution derive valid conclusions.

Why it matters: Rules of Inference appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of rules of inference in one line.
- A small example showing rules of inference in practice.
- Common mistakes students make and how to avoid them.

4. Set Theory and Relations

Relations can be reflexive, symmetric, antisymmetric and transitive; equivalence relations partition sets.

Why it matters: Set Theory and Relations appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of set theory and relations in one line.
- A small example showing set theory and relations in practice.
- Common mistakes students make and how to avoid them.

5. Functions

Functions may be injective, surjective or bijective; bijections have inverses.

Why it matters: Functions appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of functions in one line.
- A small example showing functions in practice.
- Common mistakes students make and how to avoid them.

6. Mathematical Induction

Induction proves a statement for all natural numbers via a base case and an inductive step.

Why it matters: Mathematical Induction appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of mathematical induction in one line.
- A small example showing mathematical induction in practice.
- Common mistakes students make and how to avoid them.

Quick revision

[Propositional Logic](#) · [Predicate Logic](#) · [Rules of Inference](#) · [Set Theory and Relations](#) · [Functions](#) · [Mathematical Induction](#)